Probability of Single Events

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS probability *(tūponotanga)* · outcome *(putanga)* · event *(takahanga)* · equally likely *(tūpono ōrite)*

A The probability scale ●○○

Remember: A probability is a number from 0 (impossible) to 1 (certain). An even chance is $\frac{1}{2}$.

1. Choose the best description of each event: impossible, unlikely, even chance, likely or certain.

a) Rolling a 7 on an ordinary dice _____

b) Getting a head when a coin is tossed _____

c) Rolling a number less than 7 on a dice _____

d) Rolling a 6 on a dice _____

e) Picking a red counter from a bag of 9 red and 1 blue _____

f) Rolling an even number on a dice _____

B Equally likely outcomes ●○○

Remember: Probability of an event = the number of ways it can happen ÷ the total number of outcomes. Give fractions in their simplest form.

2. An ordinary dice is rolled. Find the probability of rolling:

a) a 5 **b)** an even number **c)** a number greater than 4

d) a prime number **e)** a number less than 7 **f)** an 8

3. A bag holds 5 red, 3 blue and 2 green counters. One counter is taken out without looking. Find the probability that it is:

a) red **b)** blue **c)** not green **d)** yellow

4. Cards numbered 1 to 20 are shuffled and one card is picked. Find the probability that the number is:

a) a multiple of 5 **b)** a square number **c)** a prime number **d)** greater than 15

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

C The probability of not happening ●●○

> **Remember:** An event either happens or it does not, so $P(\text{not } A) = 1 - P(A)$. With percentages, subtract from 100%.

5. The probability that an event happens is given. Find the probability that it does not happen.

a) 0.3 **b)** $\frac{2}{5}$ **c)** 35%

d) 0.82 **e)** $\frac{7}{12}$ **f)** 4%

D One event or another ●●○

> **Remember:** If two events cannot both happen at the same time, add their probabilities: $P(A \text{ or } B) = P(A) + P(B)$.

6. Find the probability of getting:

a) a 2 or a 5 on a dice

b) a 1 or an even number on a dice

c) a red or a green counter from a bag holding 5 red, 3 blue and 2 green counters

d) a number less than 3 or greater than 15 from cards numbered 1 to 20

E Expected number ●●○

> **Remember:** Expected number = probability × number of trials.

7. Work out how many times you would expect each event to happen.

a) A fair dice is rolled 120 times. How many sixes would you expect?

b) A coin is tossed 250 times. How many heads would you expect?

c) The probability that a spinner lands on red is 0.35. It is spun 200 times. How many reds would you expect?

d) The probability that a bus is late is 0.08. How many late buses would you expect in 50 days?

e) A bag holds 3 blue and 5 white counters. A counter is picked and put back 400 times. How many blues would you expect?

f) The probability of winning a prize is $\frac{1}{20}$. How many prizes would you expect from 300 tickets?

F Problems ●●●

8. A box holds 24 chocolates. The probability of picking a dark chocolate is $\frac{3}{8}$. How many dark chocolates are in the box?

9. Three runners are in a race. The probability that Ana wins is 0.25 and the probability that Ben wins is 0.4. Find the probability that the third runner, Caleb, wins.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

10. A bag holds 4 red counters, 6 blue counters and some green counters. The probability of picking a green counter is $\frac{1}{3}$. How many green counters are there?

11. A spinner has 5 equal sections, and 2 of them are red. It is spun 80 times. How many times would you expect it not to land on red?

Extension

12. Bag A holds 8 counters and 3 of them are red. Bag B holds 12 counters and 5 of them are red. Which bag gives the better chance of picking a red counter? Show your working.

13. A dice is rolled. Find the probability that the number is prime or even (or both).

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

This page is blank so that the next version starts on a new sheet of paper.

Probability of Single Events

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS probability *(tūponotanga)* · outcome *(putanga)* · event *(takahanga)* · equally likely *(tūpono ōrite)*

A The probability scale ●○○

Remember: A probability is a number from 0 (impossible) to 1 (certain). An even chance is $\frac{1}{2}$.

1. Choose the best description of each event: impossible, unlikely, even chance, likely or certain.

a) Rolling a 0 on an ordinary dice _____

b) Getting a tail when a coin is tossed _____

c) Rolling a number greater than 0 on a dice _____

d) Rolling a 1 on a dice _____

e) Picking a blue counter from a bag of 8 blue and 2 red _____

f) Rolling an odd number on a dice _____

B Equally likely outcomes ●○○

Remember: Probability of an event = the number of ways it can happen ÷ the total number of outcomes. Give fractions in their simplest form.

2. An ordinary dice is rolled. Find the probability of rolling:

a) a 2 **b)** an odd number **c)** a number less than 3

d) a multiple of 3 **e)** a number greater than 0 **f)** a 9

3. A bag holds 4 red, 6 blue and 2 green counters. One counter is taken out without looking. Find the probability that it is:

a) red **b)** blue **c)** not green **d)** yellow

4. Cards numbered 1 to 24 are shuffled and one card is picked. Find the probability that the number is:

a) a multiple of 5 **b)** a square number **c)** a prime number **d)** greater than 15

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

C The probability of not happening ●●○

Remember: An event either happens or it does not, so $P(\text{not } A) = 1 - P(A)$. With percentages, subtract from 100%.

5. The probability that an event happens is given. Find the probability that it does not happen.

a) 0.4 **b)** $\frac{3}{7}$ **c)** 45%

d) 0.76 **e)** $\frac{5}{12}$ **f)** 8%

D One event or another ●●○

Remember: If two events cannot both happen at the same time, add their probabilities: $P(A \text{ or } B) = P(A) + P(B)$.

6. Find the probability of getting:

a) a 3 or a 4 on a dice

b) a 1 or an even number on a dice

c) a red or a green counter from a bag holding 4 red, 6 blue and 2 green counters

d) a number less than 4 or greater than 16 from cards numbered 1 to 20

E Expected number ●●○

Remember: Expected number = probability × number of trials.

7. Work out how many times you would expect each event to happen.

a) A fair dice is rolled 180 times. How many fours would you expect?

b) A coin is tossed 350 times. How many tails would you expect?

c) The probability that a spinner lands on blue is 0.45. It is spun 200 times. How many blues would you expect?

d) The probability that a train is late is 0.06. How many late trains would you expect in 50 days?

e) A bag holds 5 blue and 3 white counters. A counter is picked and put back 240 times. How many blues would you expect?

f) The probability of winning a prize is $\frac{1}{25}$. How many prizes would you expect from 400 tickets?

F Problems ●●●

8. A box holds 30 chocolates. The probability of picking a dark chocolate is $\frac{2}{5}$. How many dark chocolates are in the box?

9. Three runners are in a race. The probability that Ana wins is 0.3 and the probability that Ben wins is 0.45. Find the probability that the third runner, Caleb, wins.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

10. A bag holds 5 red counters, 7 blue counters and some green counters. The probability of picking a green counter is $\frac{1}{3}$. How many green counters are there?

11. A spinner has 8 equal sections, and 3 of them are red. It is spun 120 times. How many times would you expect it not to land on red?

Extension

12. Bag A holds 10 counters and 4 of them are red. Bag B holds 14 counters and 5 of them are red. Which bag gives the better chance of picking a red counter? Show your working.

13. A dice is rolled. Find the probability that the number is odd or greater than 4 (or both).

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.